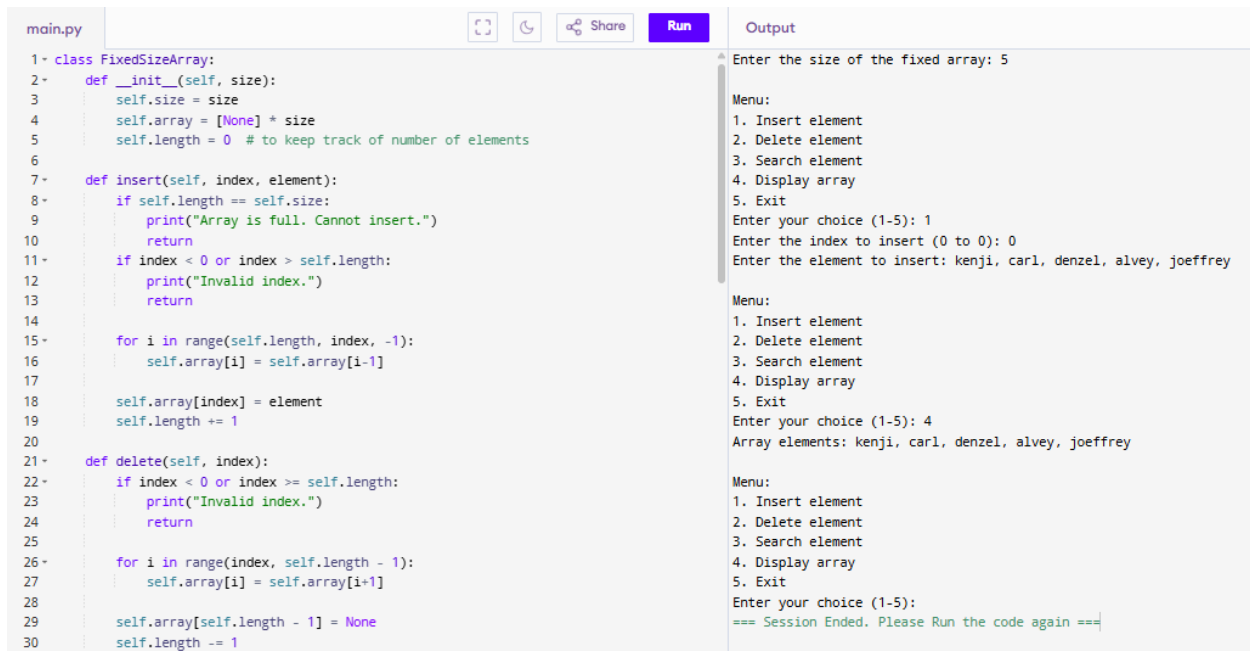


Number 1

Screenshot:



```
main.py
1 class FixedSizeArray:
2     def __init__(self, size):
3         self.size = size
4         self.array = [None] * size
5         self.length = 0 # to keep track of number of elements
6
7     def insert(self, index, element):
8         if self.length == self.size:
9             print("Array is full. Cannot insert.")
10            return
11        if index < 0 or index > self.length:
12            print("Invalid index.")
13            return
14
15        for i in range(self.length, index, -1):
16            self.array[i] = self.array[i+1]
17
18        self.array[index] = element
19        self.length += 1
20
21    def delete(self, index):
22        if index < 0 or index >= self.length:
23            print("Invalid index.")
24            return
25
26        for i in range(index, self.length - 1):
27            self.array[i] = self.array[i+1]
28
29        self.array[self.length - 1] = None
30        self.length -= 1
```

Output

```
Enter the size of the fixed array: 5
Menu:
1. Insert element
2. Delete element
3. Search element
4. Display array
5. Exit
Enter your choice (1-5): 1
Enter the index to insert (0 to 0): 0
Enter the element to insert: kenji, carl, denzel, alvey, joeffrey
Menu:
1. Insert element
2. Delete element
3. Search element
4. Display array
5. Exit
Enter your choice (1-5): 4
Array elements: kenji, carl, denzel, alvey, joeffrey
Menu:
1. Insert element
2. Delete element
3. Search element
4. Display array
5. Exit
Enter your choice (1-5):
=== Session Ended. Please Run the code again ===
```

Programiz Link: <https://www.programiz.com/online-compiler/3MShU3ZfenBna>

Code:

```
class FixedSizeArray:
```

```
    def __init__(self, size):
```

```
        self.size = size
```

```
        self.array = [None] * size
```

```
        self.length = 0 # to keep track of number of elements
```

```
    def insert(self, index, element):
```

```
        if self.length == self.size:
```

```
            print("Array is full. Cannot insert.")
```

```
            return
```

```
        if index < 0 or index > self.length:
```

```
print("Invalid index.")
```

```
return
```

```
for i in range(self.length, index, -1):
```

```
    self.array[i] = self.array[i-1]
```

```
self.array[index] = element
```

```
self.length += 1
```

```
def delete(self, index):
```

```
    if index < 0 or index >= self.length:
```

```
        print("Invalid index.")
```

```
    return
```

```
    for i in range(index, self.length - 1):
```

```
        self.array[i] = self.array[i+1]
```

```
self.array[self.length - 1] = None
```

```
self.length -= 1
```

```
def search(self, element):
```

```
    for i in range(self.length):
```

```
        if self.array[i] == element:
```

```
            return i
```

```
    return -1
```

```
def display(self):
```

```
    print("Array elements:", end=" ")
```

```
    for i in range(self.length):
```

```
        print(self.array[i], end=" ")
```

```
    print()
```

```
def main():
```

```
    size = int(input("Enter the size of the fixed array: "))
```

```
    arr = FixedSizeArray(size)
```

```
    while True:
```

```
        print("\nMenu:")
```

```
        print("1. Insert element")
```

```
        print("2. Delete element")
```

```
        print("3. Search element")
```

```
        print("4. Display array")
```

```
        print("5. Exit")
```

```
    choice = input("Enter your choice (1-5): ")
```

```
    if choice == '1':
```

```
        index = int(input(f"Enter the index to insert (0 to {arr.length}): "))
```

```
        element = input("Enter the element to insert: ")
```

```
        arr.insert(index, element)
```

```
    elif choice == '2':
```

```
        index = int(input(f"Enter the index to delete (0 to {arr.length - 1}): "))
```

```
    arr.delete(index)

elif choice == '3':

    element = input("Enter the element to search: ")

    idx = arr.search(element)

    if idx != -1:

        print(f"Element found at index {idx}.")

    else:

        print("Element not found.")

elif choice == '4':

    arr.display()

elif choice == '5':

    print("Exiting...")

    break

else:

    print("Invalid choice. Please try again.")


if __name__ == "__main__":

    main()
```